
One iteration for the convolutional dictionary update: splitting, majorization, and the placement of the channel index

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Abstract

Convolutional dictionary learning alternates a sparse-coding step with a dictionary update whose frequency-domain normal equations are diagonal plus rank K , where K is the number of training signals. The dictionary updates in use — an equality-constrained ADMM splitting solved by iterated Sherman–Morrison or by conjugate gradient, an ADMM consensus splitting, its frequency-domain form, and an accelerated proximal-gradient update — were each derived separately, and the choice between them is usually presented as a choice between methods. We show that they are one iteration. A single proximal/majorized splitting step, parameterised by the granularity of a partition of the signal index set, a basis for that index set, a per-block majorization weight, a coupling penalty and an extrapolation, produces all of them by choice of parameter value; one driver instantiated four ways reproduces a reference implementation’s functional trajectories to 4.5×10^{-16} and its converged dictionaries to 3.4×10^{-15} . Multi-channel signals are the same iteration with the channel index placed differently: as a row of the data-fidelity operator (a shared single-channel dictionary) or as a right-hand side (a multi-channel dictionary). Two consequences are algebraic rather than empirical. The first placement makes the multi-channel problem *identical* to the single-channel problem with $K \rightarrow KC$ — bitwise, in our runs — which turns the published penalty-selection rules into an identity we verify to 8.4×10^{-15} and transfers them to that representation with K replaced by KC ; the second leaves the dictionary update’s rank at K for any C , so its cost is flat in C while its penalty is not. A change of basis on the signal index set is a gauge exactly when it does not mix blocks of the partition, which explains why the frequency-domain “3D” update is a consensus method but not the consensus method, and predicts its penalty to within the resolution of the published grid search. Finally, the interior of the majorization parameter contains an update that is not any published method: keep p of the signal blocks exact and majorize the rest, giving a diagonal-plus-rank- p system with no replicas, whose cost grows linearly rather than quadratically with the index set. Its two degenerate values collapse onto two *different* reference solvers, to 1.1×10^{-14} — the test a dispatcher cannot pass — and sweeping p locates where exactness pays: on the coupled edge of the family it buys steady per-iteration progress, and on the extrapolated edge it buys none.

1 Introduction

A convolutional sparse representation approximates a signal as a sum of a few filters convolved with sparse coefficient maps, $\mathbf{s} \approx \sum_m \mathbf{d}_m * \mathbf{x}_m$, and the filters are learned by convolutional dictionary learning (CDL): alternating minimisation of

$$\arg \min_{\{\mathbf{d}_m\}, \{\mathbf{x}_{m,k}\}} \frac{1}{2} \sum_k \left\| \sum_m \mathbf{d}_m * \mathbf{x}_{m,k} - \mathbf{s}_k \right\|_2^2 + \lambda \sum_{m,k} \|\mathbf{x}_{m,k}\|_1 \quad \text{s.t.} \quad \|\mathbf{d}_m\|_2 = 1, \quad (1)$$

over K training signals, M filters and N samples per signal [Zeiler et al., 2010, Wohlberg, 2016b]. The sparse-coding step is settled: its frequency-domain system is diagonal plus rank one and is solved in closed form by the Sherman–Morrison identity [Wohlberg, 2016b]. The dictionary update is not. Fixing the coefficients, it is a convolutional form of the method of optimal directions [Engan et al., 1999] with a constraint on the filter normalisation,

$$\arg \min_{\mathbf{d}} \frac{1}{2} \|\mathbf{A}\mathbf{d} - \mathbf{s}\|_2^2 + \iota_C(\mathbf{d}), \quad (2)$$

with A the operator that convolves the zero-padded filters with the current coefficient maps, stacked over the K signals, and ι_C the indicator of the support-plus-unit-norm constraint set. In the DFT domain (2) decouples over the N spatial frequencies into systems that are diagonal plus rank K , and the rank-one shortcut does not apply. Every dictionary update in the literature is a different way of living with that rank.

The available options were derived independently and are usually compared as methods [Garcia-Cardona and Wohlberg, 2018a]: split off the constraint and solve the rank- K system directly, by iterated Sherman–Morrison (ISM) or by conjugate gradient (CG) [Wohlberg, 2016b]; give every signal its own copy of the dictionary so that each copy’s system is rank one and couple the copies by consensus [Šorel and Šroubek, 2016, Boyd et al., 2011]; treat the K signals as one volume and diagonalise a block-circulant operator with a DFT in the extra dimension [Šorel and Šroubek, 2016]; or take gradient steps and never solve a system at all [Garcia-Cardona and Wohlberg, 2018a, Beck and Teboulle, 2009]. They perform differently, and the differences are understood in terms of the structure of the linear system each solves [Garcia-Cardona and Wohlberg, 2018a]. But a list of methods, however well explained, answers only the questions someone thought to ask. It cannot say which combinations of the underlying choices were never tried, which of the methods are the same method in disguise, or what happens between them.

This paper writes the underlying choices down. Section 3 gives a single proximal/majorized splitting step for (2), derived once, whose parameters are

- Π a partition of the *signal index set* $\mathcal{I}$ (the rows of A) into blocks, each carrying one replica of the dictionary;
- Γ a change of basis on that index set;
- θ a majorization weight *per block*: keep a block’s quadratic exactly or replace it by a linear expansion;
- σ the coupling penalty, whose vanishing removes the dual variable;
- β an extrapolation of the constrained iterate.

The known updates are parameter values, not branches (Section 4): one driver, instantiated four ways, tracks a reference implementation of each of the four to relative deviation 4.5×10^{-16} in the functional and 3.4×10^{-15} in the converged dictionary. Multi-channel signals are the same iteration with the channel index placed in one of two slots (Section 5), and where it goes decides which subproblem carries the rank and which penalty rule inherits the channel count. The interior of the parameters contains working methods that are not on the list (Section 6), and different parameter values are sometimes the same iteration in disguise (Section 7). Sections 8 and 9 give the cost model that follows from the linear-system structure, the resulting selection map, and the places where the abstraction does not reach.

Concretely, we contribute: (i) a single derivation from which the four dictionary updates in use follow by parameter choice, validated against a reference implementation to machine precision for the three exact-solve families and to 2.2×10^{-9} for the iterative one; (ii) the two multi-channel representations as parameter values of the same iteration, run to convergence on colour data, together with a proof that one of them is the single-channel problem with $K \rightarrow KC$ and a confirmed prediction of the multi-channel penalty rules that follows; (iii) a gauge principle for the basis parameter that turns the known “3D = consensus in the DFT domain” observation into an exact statement with a penalty correspondence, and identifies which of its multi-channel analogues are new methods and which are re-parameterisations; (iv) a new update — partial-rank majorization — validated by a two-sided collapse onto two different reference solvers, with a measured cost model and a characterisation of where in the family it helps; and (v) a cost model and selection map over (K, C) that exploits the exchange of roles between the signal count and the channel count.

2 The dictionary update and what is held fixed

We keep every ingredient of (1) fixed and vary only the dictionary update. The functional is (1) with $\lambda = 0.1$ throughout; the constraint set is $C = \{\mathbf{x} : (I - P)\mathbf{x} = 0, \|\mathbf{x}\|_2 = 1\}$ with P the projection onto the desired filter support, whose proximal map restricts to the support and normalises; the alternating structure is auxiliary-variable coupling with a single inner iteration of each subproblem before alternating [Garcia-Cardona and Wohlberg, 2017]; and the sparse-coding stage is the standard ADMM solver for convolutional basis pursuit denoising [Wohlberg, 2016b], used unmodified from SPORCO [Wohlberg, 2017]. Reported functional values are always evaluated with the constrained (unit-norm) dictionary and the soft-thresholded coefficient maps.

Write $\mathcal{I}$ for the *signal index set*: the index set over which the rows of A in (2) are stacked. For single-channel training images $\mathcal{I} = \{1, \dots, K\}$; Section 5 shows that in one multi-channel representation it becomes the set of image-channel pairs. For $i \in \mathcal{I}$ let A_i be the corresponding row block, so $\frac{1}{2}\|A\mathbf{d} - \mathbf{s}\|^2 = \sum_{i \in \mathcal{I}} \frac{1}{2}\|A_i\mathbf{d} - \mathbf{s}_i\|^2$. In the DFT domain each A_i is, at each spatial frequency, a $1 \times M$ row, so $A^H A$ is a sum of $|\mathcal{I}|$ rank-one terms: the rank of the dictionary-update system is $|\mathcal{I}|$, whatever that index set happens to contain.

3 One iteration

Master problem. Let $\Pi = \{B_1, \dots, B_G\}$ partition $\mathcal{I}$, and give block r its own copy $\mathbf{v}_r$ of the dictionary, tied to a common $\mathbf{g}$ through a replication map $R\mathbf{g} = (w_1\mathbf{g}, \dots, w_G\mathbf{g})$ with weights $w \in \mathbb{C}^G$. Then (2) is

$$\arg \min_{\mathbf{v}, \mathbf{g}} \sum_{r=1}^G \frac{1}{2} \|A_{B_r} \mathbf{v}_r - \mathbf{s}_{B_r}\|_2^2 + \iota_C(\mathbf{g}) \quad \text{s.t.} \quad \mathbf{v} = R\mathbf{g}, \quad (3)$$

for every Π and every R with $\sum_r |w_r|^2 > 0$: the constraint forces all replicas to be the same dictionary and the block sum reassembles the residual. Applying a change of basis Γ to the signal index set — replacing $(A, \mathbf{s})$ by $(\Gamma A, \Gamma \mathbf{s})$ — leaves the minimiser unchanged whenever $\Gamma^H \Gamma \propto I$, and gives a further family of instances.

Iteration. We solve (3) by ADMM in which the $\mathbf{v}$ subproblem carries a variable metric. Assign each $i \in \mathcal{I}$ a majorization weight $\theta_i \in \{0, \theta\}$ with $\theta \in [0, 1]$, and majorize the weighted part of each block's quadratic at a point $\tilde{\mathbf{v}}$: whenever $LI \succeq \sum_{i: \theta_i > 0} A_i^H A_i$,

$$\sum_i \frac{\theta_i}{2} \|A_i \mathbf{v} - \mathbf{s}_i\|^2 \leq \sum_i \theta_i \left[\frac{1}{2} \|A_i \tilde{\mathbf{v}} - \mathbf{s}_i\|^2 + \langle A_i^H (A_i \tilde{\mathbf{v}} - \mathbf{s}_i), \mathbf{v} - \tilde{\mathbf{v}} \rangle \right] + \frac{\theta L}{2} \|\mathbf{v} - \tilde{\mathbf{v}}\|^2, \quad (4)$$

with equality at $\mathbf{v} = \tilde{\mathbf{v}}$ [Hunter and Lange, 2004]: L bounds the curvature of the majorized part only, so it shrinks as more blocks are kept exact. Minimising the resulting surrogate augmented Lagrangian gives the iteration

$$\mathbf{v}_r^{(t+1)} = \arg \min_{\mathbf{v}} \sum_{i \in B_r} \frac{1 - \theta_i}{2} \|A_i \mathbf{v} - \mathbf{s}_i\|^2 + Q_r(\mathbf{v}; \tilde{\mathbf{v}}^{(t)}) + \frac{\sigma}{2} \|\mathbf{v} - w_r \mathbf{g}^{(t)} + \mathbf{h}_r^{(t)}\|^2, \quad (5)$$

$$\mathbf{g}^{(t+1)} = \text{prox}_{\iota_C} \left(\left(\sum_r |w_r|^2 \right)^{-1} \sum_r \overline{w_r} (\mathbf{v}_r^{(t+1)} + \mathbf{h}_r^{(t)}) \right), \quad (6)$$

$$\mathbf{h}_r^{(t+1)} = \mathbf{h}_r^{(t)} + \mathbf{v}_r^{(t+1)} - w_r \mathbf{g}^{(t+1)}, \quad \tilde{\mathbf{v}}^{(t+1)} = \mathbf{g}^{(t+1)} + \beta^{(t+1)} (\mathbf{g}^{(t+1)} - \mathbf{g}^{(t)}), \quad (7)$$

where Q_r is the right-hand side of (4) restricted to B_r . Step (5) is the linear system

$$\begin{aligned} & \left[\sum_{i \in B_r} (1 - \theta_i) A_i^H A_i + (\sigma + \theta L) I \right] \mathbf{v}_r \\ &= \sum_{i \in B_r} (1 - \theta_i) A_i^H \mathbf{s}_i + \sum_{i \in B_r} \theta_i A_i^H (\mathbf{s}_i - A_i \tilde{\mathbf{v}}) + \theta L \tilde{\mathbf{v}} + \sigma (w_r \mathbf{g} - \mathbf{h}_r), \end{aligned} \quad (8)$$

which at each spatial frequency is diagonal plus a symmetric term of rank $|\{i \in B_r : \theta_i < 1\}|$, and is purely diagonal when every $\theta_i = 1$. Step (6) is one application of the constraint prox to a weighted average of the replicas. When $\sigma = 0$ the dual variable is identically zero and (7) is vacuous.

Table 1: The dictionary updates in use are parameter values of (5)–(7). q is the block size of Π , so $q = |\mathcal{I}|$ is a single replica and $q = 1$ is one replica per signal. The rank column is the rank of the per-frequency system each replica solves.

update	q	Γ	θ	σ	β	rank	inner solve
equality constraint, ISM	$ \mathcal{I} $	I	0	> 0	0	$ \mathcal{I} $	iterated Sherman–Morrison
equality constraint, CG	$ \mathcal{I} $	I	0	> 0	0	$ \mathcal{I} $	conjugate gradient
consensus	1	I	0	> 0	0	1	Sherman–Morrison
frequency-domain (“3D”)	1	$F_{ \mathcal{I} }$	0	> 0	0	1	Sherman–Morrison
proximal gradient (FISTA)	$ \mathcal{I} $	I	1	0	Nesterov	0	diagonal

Three structural facts are worth stating because they are invisible when the methods are written separately. First, the partition and the majorization weights enter the cost of an iteration only through the rank of (8) — the partition splits the rank across replicas, the weights delete rows from it — and, for the partition, through the number of replicas that must be stored. Second, the extrapolation β acts on the iteration only through $\tilde{\mathbf{v}}$, which enters (8) multiplied by θ ; *momentum is inert unless something is majorized*, which is why acceleration is available to the gradient update and not to the exact ADMM updates. Third, θ and σ are independent: majorizing does not require abandoning the dual variable, and Section 6 uses that freedom.

Known updates as parameter values. Table 1 reads the four dictionary updates in use off (5)–(7).

The consensus row deserves a word, because it is the one place where the identification is not immediate. With $q = 1$ and $w = \mathbf{1}$, (6) is prox_{ℓ_C} of the mean of $\mathbf{v}_r + \mathbf{h}_r$, which is the standard consensus $\mathbf{g}$ step [Boyd et al., 2011]; each (8) has rank one and is solved by Sherman–Morrison. The “3D” row is the same partition in a different basis and is treated in Section 7.

4 The known updates, recovered

Protocol. All four instances are run through one driver that differs only in the values passed to (5)–(7), and each is compared against SPORCO’s own solver for that family [Wohlberg, 2017] on identical inputs: the same 4 MIRFLICKR-derived training images [Huiskes and Lew, 2008] of 64×64 pixels (selected by logged seed 12345, cropped, rescaled and highpass-filtered as in Garcia-Cardona and Wohlberg 2018a), the same random initial dictionary of 16 filters of 8×8 support (seed 54321), the same $\lambda = 0.1$, the same iteration budget of 40 iterations, and the same penalty or step. Penalties follow the published selection rules [Garcia-Cardona and Wohlberg, 2018a, Table IV]: $\rho = 2.2$ for the equality-constrained and gradient updates and 3.0 for consensus, $\sigma = 0.5K + 7 = 9.0$, $\sigma = 2.2$ for consensus, and $L = 14K = 56.0$. Following the protocol of the experiments those rules were fitted on, no relaxation and no penalty adaptation are used in either stage; the inner-solver configuration — the CG tolerance and iteration cap, and the momentum policy of the gradient update — is the reference implementation’s own default.

Result. Table 3 and Figure 1 report the outcome. All four instances track the reference trajectory to a relative deviation of at most 4.5×10^{-16} at every logged iteration and reproduce the converged dictionary to a relative Frobenius error of at most 3.4×10^{-15} ; filter norms deviate from one by at most 3.3×10^{-16} . At the larger setting (Appendix B: $K = 8$, $N = 128^2$, $M = 32$, 100 iterations) the three exact-solve families still agree to 4.4×10^{-16} , 4.4×10^{-16} and 4.6×10^{-16} , while the conjugate-gradient instance agrees to 2.2×10^{-9} in the functional and 6.3×10^{-8} in the dictionary. The gap for CG is expected and is a property of that cell, not of the abstraction: an inexact inner solve terminates on a floating-point residual test, so a perturbation of order 10^{-16} can change the CG iteration count and produce a different — equally valid — approximate solve. The three cells whose inner solve is exact have no such freedom, and match to the last bit or two.

5 Multi-channel signals: where the channel index goes

A C -channel signal can be represented convolutionally in more than one way [Wohlberg, 2016a, Garcia-Cardona and Wohlberg, 2018b]. Two are basic: a *multi-channel dictionary* $\mathbf{d}_{c,m}$ with one

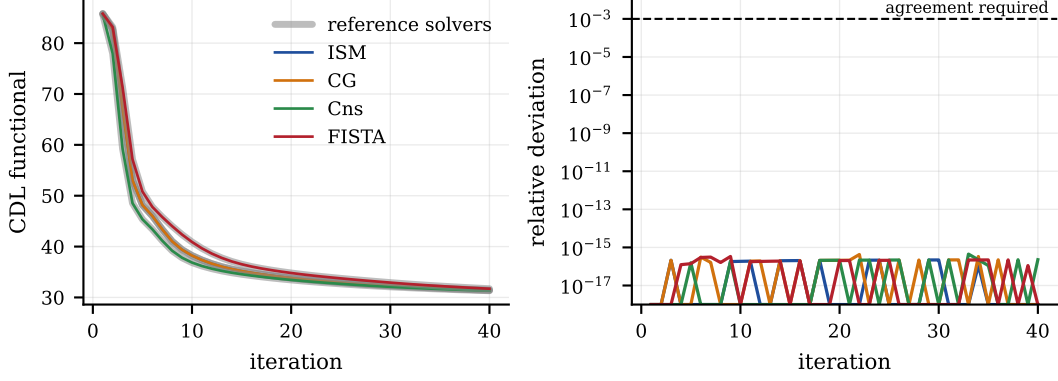


Figure 1: Left: CDL functional against iteration for the four instances (colours) over their reference solvers (thick grey). Right: relative deviation from the reference at every iteration; the dashed line is the agreement we set out to reach.

coefficient map $\mathbf{x}_{m,k}$ shared by the channels, and a *single-channel dictionary* $\mathbf{d}_m$ shared by the channels with one coefficient map $\mathbf{x}_{m,c,k}$ per channel. In the template these are not two algorithms but one iteration with the channel index in two different slots.

Single-channel dictionary: c is a row index. The dictionary update is $\arg \min_{\mathbf{d}} \frac{1}{2} \sum_{k,c} \|A_{k,c} \mathbf{d} - \mathbf{s}_{k,c}\|^2 + \iota_C(\mathbf{d})$, so the signal index set is $\mathcal{I} = \{(k, c)\}$ with $|\mathcal{I}| = KC$ and the per-frequency system has rank KC . Nothing else changes.

Multi-channel dictionary: c is a right-hand-side index. The update is $\arg \min_D \frac{1}{2} \sum_{k,c} \|A_k \mathbf{d}_c - \mathbf{s}_{k,c}\|^2 + \iota_C(D)$: the same operator A_k acts on every channel's dictionary, so $\mathcal{I} = \{k\}$, the system has rank K for every C , and the C channels are C right-hand sides sharing one factorisation. The constraint couples the channels only through the prox, which normalises each filter over its channels jointly.

Both are instances of (5)–(7) with no change to the iteration: the first changes what $\mathcal{I}$ contains, the second changes the shape of the variable the same system is applied to.

5.1 The first representation is the single-channel problem in disguise

Proposition 1 (Channel flattening). *In the single-channel-dictionary representation with plain ℓ_1 regularisation, CDL on K signals of C channels is the same instance of the template — same iterates, same functional — as CDL on the KC single-channel signals obtained by splitting the channels, for every value of $\Pi, \Gamma, \theta, \sigma, \beta$.*

Proof. The data-fidelity term is $\frac{1}{2} \sum_{k,c} \|A_{k,c} \mathbf{d} - \mathbf{s}_{k,c}\|^2$, a sum over the pairs (k, c) with no term coupling different pairs, and the regulariser $\lambda \sum_{m,k,c} \|\mathbf{x}_{m,c,k}\|_1$ is likewise a sum over pairs; the constraint acts on $\mathbf{d}$ alone. Relabelling $(k, c) \mapsto j \in \{1, \dots, KC\}$ therefore maps the objective, the sparse-coding subproblem (which is separable over signals) and the dictionary-update subproblem to those of a single-channel problem with KC signals. Every step of (5)–(7) depends on the signal index only through $\mathcal{I}$ and the partition, so the iterates correspond under the relabelling. $\square$

We verify this numerically because it is the kind of claim that is easy to state and easy to get wrong in an implementation: on 4 colour images with $C = 3$ ($|\mathcal{I}| = 12$), the multi-channel run and the flattened single-channel run agree to the last bit: the largest relative deviation of the functional over all iterations, and of the converged dictionary, is 0 for all three of the exact, consensus and gradient instances (Table 6).

The consequence is not cosmetic, because it settles how the algorithm parameters should be set. The penalty-selection rules for CDL are stated as functions of the signal count: the optimal penalty of the equality-constrained update grows like K , that of consensus does not, and the optimal gradient

Table 2: The penalty-selection rules as an identity. Replicating the training set r times and rescaling the penalty and step by the factors the template prescribes reproduces the original run exactly; omitting the rescaling does not. Deviation is the largest relative deviation of J_r/r from J_1 over 40 iterations at $K = 4$, $N = 64^2$, $M = 16$, $\sigma = 1.0$, $L = 15.0$.

instance	σ fac.	L fac.	dev. ($r=2$)	dev. ($r=3$)	dev. unrescaled
equality constraint	r	1	1×10^{-15}	1.6×10^{-15}	1.9×10^{-2}
consensus	1	1	1.4×10^{-15}	1.3×10^{-15}	(identical run)
proximal gradient	no σ	r	9.6×10^{-16}	1.1×10^{-15}	6.2×10^{-1}
partial rank (new)	r	r	1.1×10^{-15}	9.1×10^{-16}	6.1×10^{-2}

step grows like K [Garcia-Cardona and Wohlberg, 2018a, §VII-G]. Those rules are usually read as empirical fits, but read through the template they are an identity. Replicating the training set r times multiplies $|T|$ by r and multiplies every $A_i^H A_i$ term in (8) by r , so the r -fold run at $(r\sigma, rL)$ has exactly the iterates of the one-fold run at (σ, L) , and exactly r times its functional. Table 2 checks this as an identity rather than by grid search: the deviation is at most 8.4×10^{-15} for every instance, against at least 1.4×10^{-2} if the rescaling is omitted. Consensus is the case $r\sigma = \sigma$, so for it the rescaled and unrescaled runs are the same run — which is exactly why its optimal penalty does not move with the signal count.

Three things follow. First, the same identity holds for the partial-rank update of Section 6.1, with both its penalty and its step scaled: the template supplies the parameter rule for a method that has never been run. Second, by Proposition 1 the identity applies to the multi-channel problem with $r = C$: with C replicated channels, the single-channel-dictionary run at $(C\sigma, CL)$ reproduces the single-channel run exactly (deviation at most 1.6×10^{-15}), so the published rules transfer to that representation with K replaced by KC — not with K , which is how the published colour experiment sets them. Third, the identity is exact only for replicated data; on real colour images the channels are correlated but not identical, so it becomes a prediction, which Figure 2 tests.

On real colour data the two representations behave as the rank statement leads one to expect, but not for the reason one might assume. For the single-channel dictionary the grid-searched optimal gradient step grows by $1.94\times$ and $3.10\times$ at $C = 2, 3$, against the $2\times$ and $3\times$ that $K \rightarrow KC$ predicts. For the multi-channel dictionary, whose system stays rank K whatever C is, the optimum nevertheless grows, by $2.32\times$ and $3.55\times$. The measured curvature of the data term explains both: with the same images repeated it scales exactly with r (425.0 and 637.5 against 425.0 at $r=1$: factors 2.00 and 3.00), and on real colour it grows by $2.25\times$ and $2.95\times$ for the single-channel dictionary and by $1.41\times$ and $3.52\times$ for the multi-channel one. In the first representation the curvature grows because the channel index adds rows; in the second it grows because one shared coefficient map has to explain C channels at a fixed sparsity weight, so the map itself grows. The rank statement is about cost, then, and the penalty statement is about curvature; they coincide only in the representation where the channel index enters the rank. That distinction is a concrete prediction of a fact the published colour experiment reports as a fudge: its gradient step had to be multiplied by three — the number of channels — “for a more stable convergence”.

5.2 Both representations, run on colour

Table 3 and Figure 3 report both representations at the scale of the published colour experiment ($M = 64$ filters of $8 \times 8 \times 3$ or 8×8 support, 4 colour images of 256×256 pixels, $\lambda = 0.1$, 100 iterations), each instantiated three ways and each against a matched reference run of SPORCO’s own CDL on the corresponding path at the same λ . Every run reduces the multi-channel functional to at most 0.279 (multi-channel dictionary) and 0.327 (single-channel dictionary) of its initial value, and agrees with its reference to 2×10^{-13} and 1.3×10^{-14} respectively. The dictionaries the two representations learn are shown in Appendix D.

The cost difference between the two representations separates into two effects. Both stages have to transform a coefficient-map array that is C times larger in the single-channel-dictionary representation, which is a cost every cell pays; on top of that, only the exact solve pays for the rank. Measured, the exact-solve instance takes $1.7\times$ longer in that representation than in the multi-channel one while the gradient instance — which forms no system — takes $1.3\times$ longer; the common factor is the array

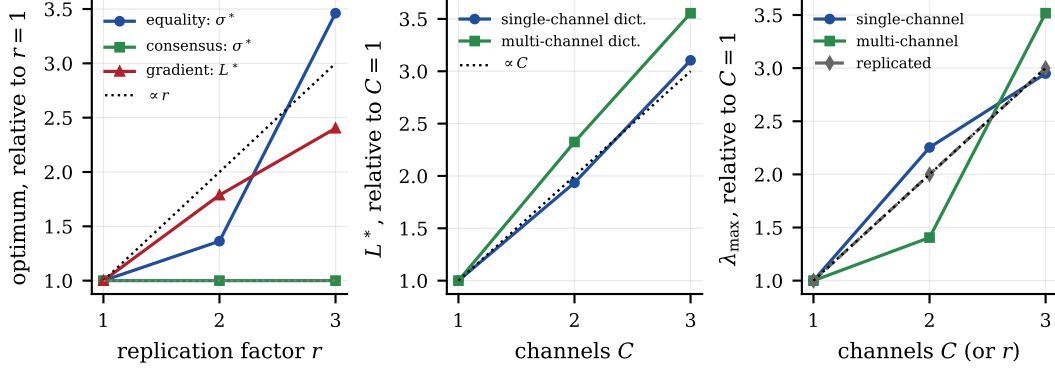


Figure 2: Grid-searched optimum of the dictionary-update penalty (left, centre) and gradient step (right) against the number of channels, for the two representations, with the $\propto C$ and constant references (dotted). The optimum moves with C in both representations; only in the single-channel-dictionary representation is that because the channel index enters the rank.

Table 3: Each update and each representation as a value of the template’s parameters, with its deviation from the matched reference run and the functional it converges to. Deviations are the largest relative deviation of the CDL functional over all logged iterations. The two interior cells have no external reference, so the entry reports the worst deviation of their two degenerate values from the reference solvers they collapse to. Functional values are comparable within a block of rows, not across blocks, since the blocks differ in problem size.

update	representation	parameter values	deviation from reference	converged functional
equality constraint, ISM	single-channel	$q= \mathcal{I} , \theta=0, \sigma=9$	2.2×10^{-16}	31.4033
equality constraint, CG	single-channel	$q= \mathcal{I} , \theta=0, \sigma=9, \text{CG}$	4.3×10^{-16}	31.4112
consensus	single-channel	$q=1, \theta=0, \sigma=2.2$	4.5×10^{-16}	31.3246
proximal gradient	single-channel	$q= \mathcal{I} , \theta=1, \sigma=0, \beta_N$	3.4×10^{-16}	31.7143
equality constraint, ISM	multi-channel dictionary	$q= \mathcal{I} , \theta=0, \sigma=9$	7.9×10^{-16}	444.745
consensus	multi-channel dictionary	$q=1, \theta=0, \sigma=2.2$	1.1×10^{-15}	446.770
proximal gradient	multi-channel dictionary	$q= \mathcal{I} , \theta=1, \sigma=0, \beta_N$	2×10^{-13}	505.927
equality constraint, ISM	single-channel dictionary	$q= \mathcal{I} , \theta=0, \sigma=13$	7.4×10^{-16}	591.927
consensus	single-channel dictionary	$q=1, \theta=0, \sigma=2.2$	4.1×10^{-16}	586.776
proximal gradient	single-channel dictionary	$q= \mathcal{I} , \theta=1, \sigma=0, \beta_N$	1.3×10^{-14}	593.117
partial rank, $p=2$ (new)	single-channel	$q= \mathcal{I} , \theta=1, p=2, \sigma=2.15$	collapse 3.9×10^{-16}	346.1514
block consensus, $q=8$	single-channel	$q=8, \theta=0, \sigma=2.2$	collapse 3.7×10^{-16}	303.9645
partial rank, $p=12$ (new)	single-channel dict.	$q= \mathcal{I} , \theta=1, p=12, \sigma=4.64$	collapse 1.5×10^{-15}	337.9909
partial rank, $p=1$, extrapolated (new)	single-channel	$q= \mathcal{I} , \theta=1, p=1, \sigma=0, \beta_N$	collapse 3.6×10^{-16}	306.4700

size and the difference between the two is the rank. The rank effect on its own is isolated in Figure 8, where only $|\mathcal{I}|$ varies.

5.3 The third representation costs the template nothing

A third multi-channel representation couples the per-channel coefficient maps of an image by an inter-channel joint-sparsity term [Wohlberg, 2016a]. In the template it is not a new dictionary update at all: the coupling is a term of the sparse-coding subproblem, so the dictionary-update instance is the one the single-channel-dictionary representation already uses, with the same signal index set of image–channel pairs. We ran it to check exactly that. With $K = 4$ colour images, $M = 32$ filters and a coupling weight $\mu = 0.1$, all four instances — the two exact ones, the gradient one and the

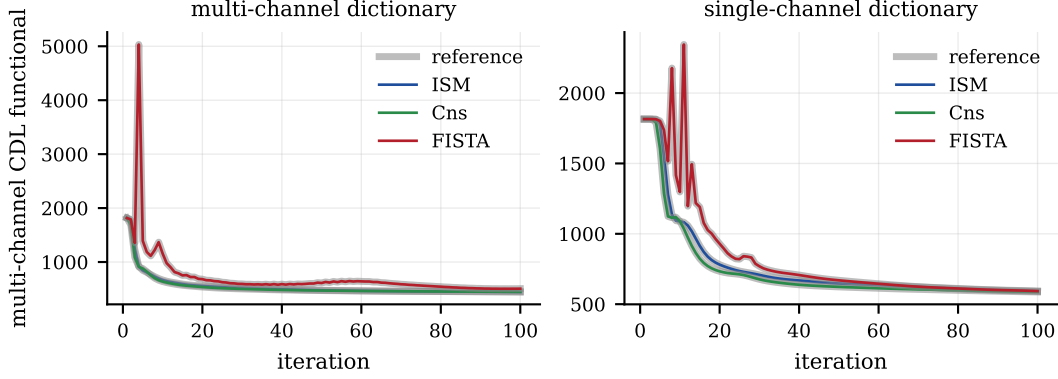


Figure 3: Colour dictionary learning from the template, both representations, each instance over its matched reference run (thick grey). The gradient instance’s initial transient — noted in the original comparison — is reproduced too, which is the kind of detail a wrapper would not have to get right.

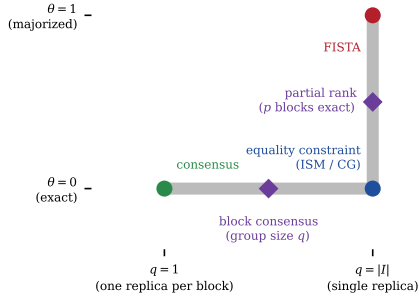


Figure 4: The two interiors. Along the bottom edge the partition granularity interpolates between consensus and the equality-constrained update, with block consensus in between (known: Choudhury et al. 2017). Along the right edge the majorization weight interpolates between the equality-constrained update and the gradient update; its interior — keeping p blocks exact — is a family of updates that is not on the list. Momentum is available only on the majorized edge.

partial-rank one of Section 6.1 — run unchanged and reduce that representation’s functional to at most 0.511 of its initial value in 100 iterations. Its values are not comparable with those reported elsewhere, because the coupling changes the functional being minimised; that is the point, and it is why this representation earns no separate entry in Table 3.

It also gives Proposition 1 a negative control. The joint-sparsity term couples the channels of an image, so the problem is no longer separable over image–channel pairs and the flattening identity should fail. It does: the same instance run on the multi-channel problem and on its channel-flattened counterpart now differ by 5.5×10^{-1} in the functional and 5.8×10^{-1} in the dictionary, against exactly zero without the coupling.

6 The interior

A template earns its name by containing methods nobody wrote down. Two of its parameters have interiors: the partition Π can be coarser than singletons and finer than the whole index set, and the majorization weights can be 0 on some blocks and 1 on others. Figure 4 places the known updates and the two interiors.

6.1 Partial-rank majorization

Set $\theta_i = 0$ on a subset $E \subset \mathcal{I}$ of p blocks and $\theta_i = 1$ on the rest, with a single replica ($q = |\mathcal{I}|$). Then (8) becomes

$$\left[\sum_{i \in E} A_i^H A_i + (\sigma + L)I \right] \mathbf{v} = \sum_{i \in E} A_i^H \mathbf{s}_i + \sum_{i \notin E} A_i^H (\mathbf{s}_i - A_i \tilde{\mathbf{v}}) + L \tilde{\mathbf{v}} + \sigma(\mathbf{g} - \mathbf{h}) : \quad (9)$$

System (9) is diagonal plus rank p , solved exactly by p steps of the Sherman–Morrison recursion, and the blocks outside E appear in it only through a gradient. No replicas are created, so memory is that of the equality-constrained update; the per-frequency solve costs $O(p^2)$ instead of $O(|\mathcal{I}|^2)$; and

Table 4: The family laid out. Rows are values of the parameters of (5)–(7); the last column says whether the resulting update is one already in use, one that is a re-parameterisation of another row, or one that is not in the literature we are aware of. F_κ is an unnormalised DFT on the signal index set with $F_\kappa^H F_\kappa = \kappa I$.

q	Γ	θ, p	σ	β	status
$ \mathcal{I} $	I	$\theta=0$	> 0	0	equality-constrained update (ISM or CG inner solve)
1	I	$\theta=0$	> 0	0	consensus update
1	F_κ	$\theta=0$	> 0	0	frequency-domain (“3D”) update: same q , other Γ
$ \mathcal{I} $	I	$\theta=1, p=0$	0	Nesterov	proximal-gradient update
$ \mathcal{I} $	F_κ	any	> 0	any	re-parameterisation of the equality constraint (Prop. 2)
1	F_κ	$\theta=0$	> 0	0	not a re-parameterisation: F_κ mixes blocks (§7)
$1 < q < \mathcal{I} $	I	$\theta=0$	> 0	0	block consensus [Choudhury et al., 2017, Boyd et al., 2011]
$ \mathcal{I} $	I	$\theta=1, 0 < p < \mathcal{I} $	> 0	0	partial rank (§6.1)
$ \mathcal{I} $	I	$\theta=1, p=0$	> 0	0	linearised ADMM: a standard device, not reported for this update
$ \mathcal{I} $	I	$0 < \theta < 1$	> 0	0	partial linearisation: no rank reduction, so of no use
$ \mathcal{I} $	I	$\theta=1, p > 0$	> 0	Nesterov	extrapolated partial rank: runs, but with no guarantee
spatial tiling; masked CDL; spatial-domain updates					outside the family (§9)

Table 5: Curvature of the majorized part, $\lambda_{\max}$ over all frequencies, against the number p of blocks kept exact, measured on one fixed state ($K = 16$, $N = 128^2$, $M = 32$, after 20 iterations) so that every p sees the same coefficient maps. Both orderings reduce it; neither dominates.

p	0	1	4	8	12	15	16
largest coefficient-map energy first	1945	1939	1565	1421	240	42	0
arbitrary subset	1945	1828	1435	560	309	80	0

p is free. At $p = |\mathcal{I}|$ this is the equality-constrained update and at $p = 0$ (with $\sigma = 0$ and Nesterov β) it is the gradient update, so the family joins two *different* published methods through its interior.

Two design questions come with the cell. What step to use: the majorization is valid for any $L \geq \lambda_{\max}$ of the *majorized* part, which is the largest eigenvalue of a $(|\mathcal{I}| - p) \times (|\mathcal{I}| - p)$ Gram matrix per frequency and is cheap to compute exactly, so the cell can be run with no step-size parameter at all; we report both that parameter-free variant and the published rule read through the template, $L = 14(|\mathcal{I}| - p)$. And which blocks to keep exact: we order them by coefficient-map energy $\|\hat{A}_i\|_F$, recomputed each outer iteration, on the intuition that removing the largest blocks removes the top of the spectrum and licenses a larger step. Measured on a fixed state (Table 5), the first half of that intuition holds — the curvature of the majorized part falls steeply with p , from 1945 at $p = 0$ to 1421 (27% lower) at $p = |\mathcal{I}|/2$ for that ordering and to 560 (71% lower) for an arbitrary subset — but the second half does not: ordering by energy is *not* uniformly better than taking an arbitrary subset, and at $p = |\mathcal{I}|/2$ it is $2.54\times$ worse, because the top eigenvector of the Gram matrix is spread over correlated blocks rather than concentrated on the largest one. All the runs below use the energy ordering, so they are if anything handicapped; choosing the subset to minimise the remaining curvature is an open question the family poses and we do not answer.

Consistency. There is no external reference for $0 < p < |\mathcal{I}|$, so the cell is checked at its two degenerate values, through the same code path with only p changed: $p = |\mathcal{I}|$ against the reference iterated-Sherman–Morrison solver, and $p = 0$ with $\sigma = 0$ and Nesterov extrapolation against the reference proximal-gradient solver. The worst deviation over both checks — functional trajectory and converged dictionary — is 1.1×10^{-14} (Table 6). A dispatcher cannot do this: it has no value of p between its branches.

Behaviour, and where it helps. Figures 5 and 6 report sweeps at $K = 16$, $N = 128^2$, $M = 32$ ($|\mathcal{I}| = 16$) over $p \in \{0, 1, 2, 4, 8, 12, 15, 16\}$, on three edges of the family: with a dual variable and no extrapolation, without a dual variable and with it, and with both. The penalty, where there is one, is tuned once at the exact endpoint $p = |\mathcal{I}|$, which if anything favours that endpoint; times are dictionary-update times, since the sparse-coding stage is identical across cells and adds the same constant (0.625 s per iteration here) to every curve.

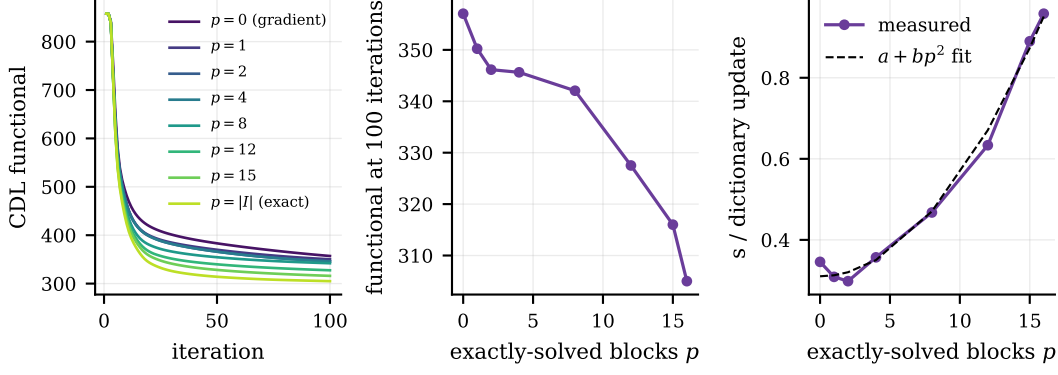


Figure 5: Partial-rank majorization. Left: functional against iteration as the number p of exactly-treated blocks grows. Centre: functional after 100 iterations against p . Right: measured cost of one dictionary update against p , with the $a + bp^2$ form the Sherman–Morrison recursion predicts.

Per iteration, exactness simply buys progress: on the coupled edge with the parameter-free step the functional after 100 iterations falls monotonically from 357.0392 at $p = 0$ to 305.0088 at $p = |I|$, and the cost per update rises from 0.3458 s to 0.9581 s. Per second, the answer depends on the budget (Figure 7). Given 10 s in the dictionary update the best member of the family is interior, $p = 2$, reaching 380.012 against 405.179 for keeping nothing exact and 423.525 for keeping everything exact: at a short budget the interior beats both of the methods it joins. Given 96 s the exact endpoint has caught up and wins, at 305.009. So the interior is worth having exactly when iterations are expensive relative to the accuracy wanted — which is the regime the rank- $|I|$ solve was a problem in to begin with.

The comparison that matters most is against the proximal-gradient corner, and there the honest answer is that the interior does not win: that corner reaches 307.668 at the short budget and 306.492 at the long one, better than every point of the coupled edge except its exact endpoint at the largest budget. The family explains why, and the explanation is structural rather than empirical: the extrapolation parameter is inert unless something is majorized (Section 3), so full majorization is the only setting at which acceleration is available at all, and here acceleration is worth more than exactness.

The two edges that use the published step rule sharpen that point, because on both of them the interior is not where the best member is. With a dual variable the functional improves monotonically with p , from 332.0393 at $p = 0$ to 305.0089 at $p = 16$ — that is, all the way to the exact endpoint, with nothing gained in between. Without a dual variable — where $p = 0$ is the proximal-gradient update — it is flat at 306.4915 and then drifts upwards. The reason is a finding about the published rule rather than about the cell: read through the template that rule is $L = 14(|I| - p)$, linear in the number of majorized blocks, but the measured curvature of the majorized part is not linear in that number (Table 5), so the rule mis-sets the step everywhere except at the corner it was fitted at. Exactness and step size are substitutes: making blocks exact pays when the step you are entitled to take is conservative, and not when the step you are taking is already aggressive.

Combining a dual variable with extrapolation — the one row of Table 4 for which we know of no convergence theory — is also available, and also runs: it reaches 305.0089 at $p = 16$, against 332.0393 for the same edge without extrapolation. We report it because the family contains it, not because we can vouch for it.

The same cell on multi-channel data. In the single-channel-dictionary colour representation the signal index set is the set of image–channel pairs, so it is C times larger — $|I| = 12$ at $K = 4$ colour images — which is the regime the cell is meant for. It is also the one place where the interior comes out ahead of the proximal-gradient corner outright. With the published step rule the sweep behaves as it does on single-channel data, 340.1285 at $p = 0$ rising with p ; with the parameter-free step it falls monotonically, from 347.8196 at $p = 0$ to 339.5766 at $p = 9$, which is below the 340.1285 the proximal-gradient corner reaches. The margin is small and it is per iteration, not per second: that setting costs 0.4636 s per update against 0.0826 s for the corner it beats. Appendix C plots both sweeps. It collapses correctly here too: all four degenerate values of the two interior cells — the

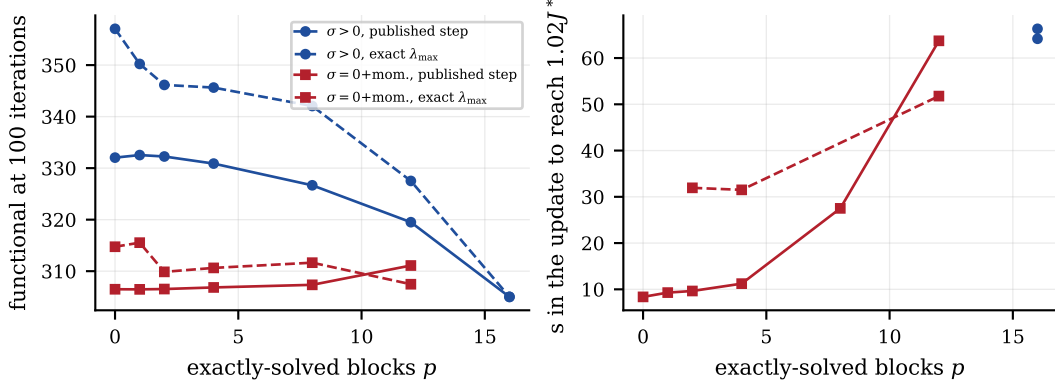


Figure 6: The majorization interior on three edges of the family and under two step policies. Left: functional after the iteration budget, against the number p of blocks kept exact; it falls with p wherever a dual variable is present and is flat without one. Right: dictionary-update time to reach a functional common to all of them, which the cheap end of each edge reaches first; a point is absent where that configuration never reached the common target inside the iteration budget, which is why the coupled edge appears only at its exact endpoint.

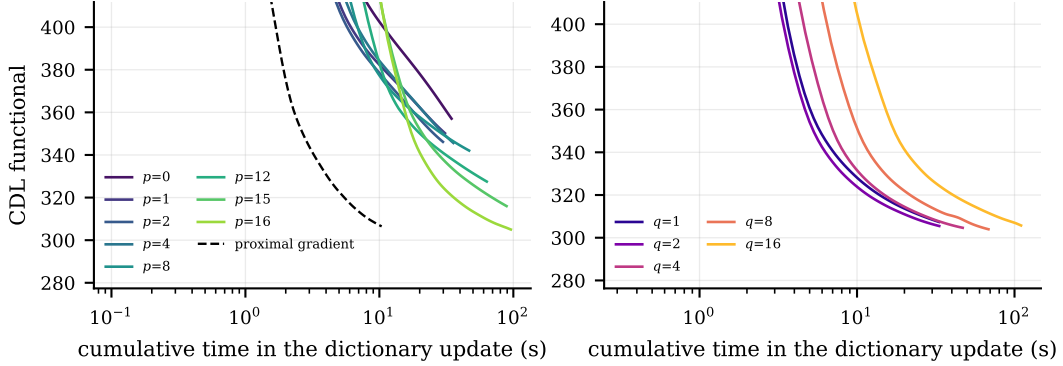


Figure 7: Dictionary-update time to reach a common target functional, against the two interior parameters: the number of exactly-treated blocks (left) and the group size of the partition (right).

two of the majorization parameter and the two of the partition — reproduce their reference solvers in this representation to at most 8.7×10^{-15} , over both the functional trajectory and the converged dictionary.

6.2 Block consensus, and why we do not claim it

The other interior — partitioning $\mathcal{I}$ into G groups of size q , one replica each, each solving a rank- q system — is a genuine cell of the template, and it is *not* new: it is the construction of Choudhury et al. [2017] for convolutional sparse coding, and in general optimisation it is the standard grouped-consensus device [Pierra, 1984, Boyd et al., 2011, Spingarn, 1985]. We run it because the template contains it and because it is the natural point of comparison for Section 6.1, and we report it with attribution. Our version does differ in one respect that the template makes natural: each group’s diagonal-plus-rank- q system is solved by the Sherman–Morrison recursion, which costs $O(q)$ per signal and forms no factorisation to store, rather than by a Cholesky factorisation of a reordered system. The two are the same order when $M \gg q$; the difference is memory and code, not asymptotics. Across $q \in \{1, 2, 4, 8, 16\}$ the functional after 100 iterations is best at $q = 8$, at 303.9645, against 307.4136 for one replica per signal (consensus) and 305.7386 for a single replica (the equality-constrained update): this interior, like the majorization interior, is not at an endpoint, and we have not seen that reported. The same two degenerate values collapse onto the reference consensus and equality-constrained solvers to 3.7×10^{-16} and 3.7×10^{-16} respectively.

Table 6: Consistency checks. Top: the channel-flattening identity of Proposition 1. Middle: the basis gauge of Proposition 2, with the control obtained by omitting the penalty rescaling. Bottom: the degenerate values of the two interior cells against the reference solver each collapses to.

check	compared with	functional deviation	dictionary deviation
flattening, exact solve	single-channel, $K \rightarrow KC$	0	0
flattening, consensus	single-channel, $K \rightarrow KC$	0	0
flattening, gradient	single-channel, $K \rightarrow KC$	0	0
DFT over images, $\kappa = 4$	untransformed, σ	3.1×10^{-16}	(control 4.5×10^{-2})
DFT over channels, $\kappa = 3$	untransformed, σ	3.1×10^{-16}	(control 3.7×10^{-2})
DFT over both, $\kappa = 12$	untransformed, σ	2.2×10^{-16}	(control 6.5×10^{-2})
partial rank, $p = \mathcal{I} $	reference ISM solver	3.7×10^{-16}	1.1×10^{-14}
partial rank, $p = 0$	reference PGM solver	3.9×10^{-16}	8.3×10^{-15}
block consensus, $q = 1$	reference consensus solver	3.7×10^{-16}	3.1×10^{-15}
block consensus, $q = \mathcal{I} $	reference ISM solver	3.7×10^{-16}	1.1×10^{-14}

7 When are two parameter values the same iteration?

Proposition 2 (Basis covariance). *Let Γ act on the signal index set with $\Gamma^H \Gamma = \kappa I$, and suppose Γ is block diagonal with respect to the partition Π . Then the template run on $(\Gamma A, \Gamma \mathbf{s})$ with penalty $\kappa \sigma$ and step κL produces exactly the iterates of the run on $(A, \mathbf{s})$ with σ and L .*

Proof. Because Γ is block diagonal with respect to Π , block r of the transformed problem is $(\Gamma_r A_{B_r}, \Gamma_r \mathbf{s}_{B_r})$ with $\Gamma_r^H \Gamma_r = \kappa I$, so its data term is $\kappa \cdot \frac{1}{2} \|A_{B_r} \mathbf{v} - \mathbf{s}_{B_r}\|^2$ and its majorizer is κ times the original one. The objective of (5) at penalty $\kappa \sigma$ and step κL is therefore κ times the original objective, with the same minimiser. Step (6) is a prox of an indicator, invariant to positive scaling of its quadratic, and (7) is linear. $\square$

So the basis is a *gauge*: a whole column of parameter values that produces no new method, only a reparameterisation of the penalty. With a single replica, every Γ with $\Gamma^H \Gamma \propto I$ qualifies. Table 6 confirms this for the unnormalised DFT over the image index ($\kappa = 4$), over the channel index ($\kappa = 3$) and over both ($\kappa = 12$) in the single-channel-dictionary representation: agreement to 3.1×10^{-16} , against 3.7×10^{-2} or worse if the penalty is not rescaled. Two consequences follow.

A “channel-3D” method is not a new method, and the version that is, is not better. Transforming the channel index by a DFT — an obvious multi-channel analogue of the frequency-domain update — is, with a single replica, the equality-constrained update at penalty $C\sigma$, exactly and not approximately. Combined with the singleton partition it is a different iteration, and one that no one has run; we ran it, at the penalty correspondence $\sigma = C\sigma_{\text{Cns}} = 6.60$, and it reaches 94.2130 against consensus’s 93.1873. Filling that cell was possible and not worthwhile, which is also something a family should be able to tell you before you implement it.

The frequency-domain (“3D”) update is a consensus method but not the consensus method. With one replica per signal, the block-DFT is *not* block diagonal with respect to the partition: it mixes signals across blocks, so Proposition 2 does not apply. It is known that the “3D” update is ADMM consensus with the data term and constraint expressed in the DFT domain [Garcia-Cardona and Wohlberg, 2018a]; the template sharpens this to: same partition value, different basis value, hence the same cell but a different iteration. Both statements are borne out. The data term of the transformed problem is κ times the original, so the two should be compared at $\sigma_{3D} = \kappa \sigma_{\text{Cns}}$; at that correspondence the two runs reach 93.1509 and 93.1873, a relative difference of 3.9×10^{-4} — close but not equal, as the mixing implies. The published grid search is consistent with the correspondence: its optimal σ for the frequency-domain update at $K = 10$ is exactly $10 \times$ its optimal σ for consensus, and within one grid step of $K \times$ at $K = 5, 20, 40$ [Garcia-Cardona and Wohlberg, 2018a, Table II].

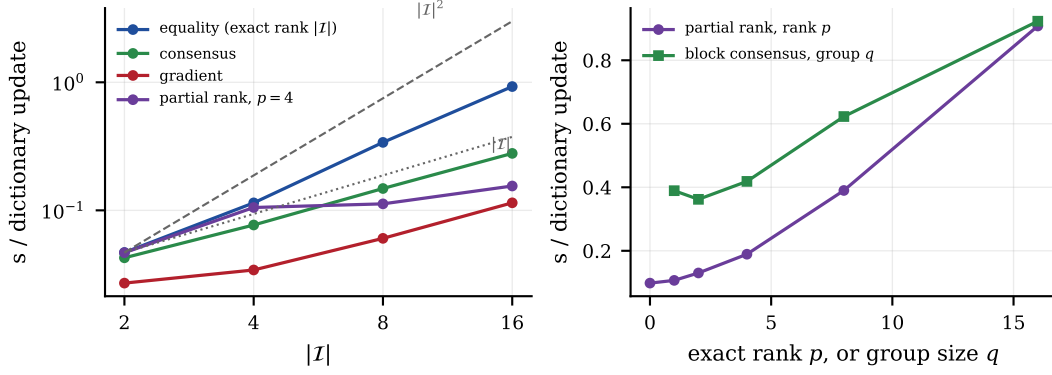


Figure 8: Cost of one dictionary update. Left: against the size of the signal index set, with linear and quadratic references. Right: against the exact rank of the two interior cells at $|\mathcal{I}| = 16$. Keeping p directions exact without replicas is cheaper, at every rank, than keeping them exact by splitting into groups of p .

8 Cost, mechanism, and which value to use

Where the cost is. One iteration costs the transforms, which are $O(|\mathcal{I}|MN \log N)$ and identical in every cell, plus the solves. With a partition into blocks of size q , the Sherman–Morrison recursion costs $O(q)$ per signal, so $O(q|\mathcal{I}|MN)$ in total: linear in q , from $O(|\mathcal{I}|MN)$ at $q = 1$ to $O(|\mathcal{I}|^2MN)$ at $q = |\mathcal{I}|$. Partial-rank majorization has one replica, so its solve costs $O(p^2MN)$ plus $O(|\mathcal{I}|MN)$ for the gradient of the majorized part. The two interiors therefore approach the same endpoints along different curves — linear in the replica size, quadratic in the exact rank — and they differ in memory: block consensus holds $|\mathcal{I}|/q$ replicas and their duals, partial-rank majorization holds one. Figure 8 measures both. The exact update grows by $19.8\times$ between $|\mathcal{I}| = 2$ and 16 — between linear and quadratic, as a transform cost that is linear plus a solve that is quadratic implies — while consensus and the gradient update grow by $6.5\times$ and $4.2\times$, i.e. linearly. Partial-rank majorization at fixed p grows by $3.3\times$: it inherits the *linear* scaling of the gradient update, because its exact rank does not grow with the index set, while retaining an exact solve on p directions. At $|\mathcal{I}| = 16$ that is $6.0\times$ cheaper than the exact update and $1.35\times$ the cost of the gradient update. Table 7 gives per-update times for every cell at a common problem size. The right panel also shows a cost the model does not include: block consensus at $q = 1$ is slightly *more* expensive than at $q = 2$, because $|\mathcal{I}|$ separate small solves carry a per-replica overhead that $|\mathcal{I}|/2$ larger ones do not.

Why the ranking is what it is. The published finding that the gradient update beats the exact update in serial, while consensus wins when parallel [Garcia-Cardona and Wohlberg, 2018a], reads off the same structure: the gradient update pays $O(|\mathcal{I}|MN)$ and buys no exactness; the exact update pays $O(|\mathcal{I}|^2MN)$ and buys the fastest per-iteration progress; consensus pays $O(|\mathcal{I}|MN)$ and buys exactness per replica at the price of consensus lag, which is why it needs the replicas to run concurrently to pay off. Partial-rank majorization is the observation that this is a continuum, and that the endpoints are not the best point on it.

Selection map. The rank of the dictionary-update system is $|\mathcal{I}|$ and the rank of the sparse-coding system is the number of dictionary channels; the two representations of Section 5 distribute C between them. In the single-channel-dictionary representation the dictionary update carries rank KC and the sparse coding stays rank one; in the multi-channel-dictionary representation the dictionary update stays rank K and the sparse coding carries rank C [Garcia-Cardona and Wohlberg, 2018a]. The channel count is conserved: it inflates exactly one of the two subproblems, and the representation decides which. Table 8 is the resulting rule. For a multi-channel dictionary and small C (colour), the dictionary update is insensitive to C and any cell will do; the sparse-coding stage is the one to worry about. For a shared single-channel dictionary, or for many channels, $|\mathcal{I}| = KC$ is large, the exact solve is quadratic in it, and the useful cells are those that keep the rank low: consensus if replicas can run concurrently, partial-rank majorization if not. Our measurements put the crossover between $p = |\mathcal{I}|$ and interior p at $|\mathcal{I}| = 16$ already. The duality also runs the other way: the sparse-coding

Table 7: Measured cost of one dictionary update at a common problem size ($|I| = 16$, $N = 128^2$, $M = 32$), median of five timings after warm-up, single-threaded transforms on dedicated cores. Absolute times are machine-specific; the comparison across cells is what the cost model predicts.

cell	equality, ISM	equality, CG	consensus	gradient	partial rank, $p=4$	block consensus, $q=4$
s / update	0.9264	0.2036	0.3023	0.1335	0.1892	0.4186

Table 8: Which cell to use where. The rank of the dictionary-update system is the size of the signal index set; the rank of the sparse-coding system is the number of dictionary channels. The representation decides which of the two subproblems the channel count inflates.

regime	update rank	coding rank	cell to use
single channel, small K	K	1	any; the exact solve is cheap
single channel, large K	K	1	partial rank or the gradient update in serial; consensus in parallel
multi-channel dictionary	K	C	as above: C does not enter the update, and the sparse-coding stage is the constraint
single-channel dictionary	KC	1	partial rank or consensus; the exact solve is quadratic in KC
many channels, e.g. spectral	K or KC	C or 1	choose the representation to put the rank where you can afford it

subproblem has the same diagonal-plus-low-rank structure with C in the role of $|I|$, so the same cells — including partial rank — are available there, and are worth having exactly when C is large.

9 Where the abstraction stops

Spatial tiling is not a basis. Tiling the K images into one large image so that the update has $K' = 1$ [Šorel and Šroubek, 2016] looks like the frequency-domain update: both map many signals to one. But tiling is not a map Γ on the signal index set with $\Gamma^H \Gamma \propto I$. Circular convolution on the tiled image wraps between adjacent tiles, whereas the original problem wraps within each image, so the tiled problem has a different data-fidelity term — not a different basis for the same one. It is consistent with this that the published grid search finds a different optimal penalty for tiling than for the equality-constrained update at the same K [Garcia-Cardona and Wohlberg, 2018a, Table II]. We therefore do not claim tiling as an instance; it sits outside the template, and a template that pretended otherwise would be wrong.

What is not parameterised. The template covers the dictionary update only. The sparse-coding stage and the coupling between the stages are held fixed, and although the same replicate-and-majorize construction plainly applies to the sparse-coding subproblem (where the rank is the channel count), we have not carried it there. The masked variants of CDL [Heide et al., 2015, Garcia-Cardona and Wohlberg, 2018a] require a second splitting of the data-fidelity term through the mask; that is an additional constraint block, not a value of any parameter here. Inter-channel joint sparsity [Wohlberg, 2016a] is expressed in this framework as the single-channel-dictionary representation with a different sparse-coding prox, so its distinguishing content lies in the stage we hold fixed; Proposition 1 does *not* apply to it, because the joint-sparsity term couples the channels of a pair. Spatial-domain dictionary updates [Pappayan et al., 2017] are not frequency-domain solves of (2) at all and are outside the family. Recent primal-dual solvers with no penalty parameter are likewise outside it.

Guarantees. Convergence of the interior cells inherits what is known for proximal ADMM with a positive semidefinite metric [Deng and Yin, 2016] and for multi-block consensus [Deng et al., 2017], provided $L \geq \lambda_{\max}$ of the majorized part — which the published step-size rule does not enforce: at $p = 0$ we measure the exact $\lambda_{\max}$ of the majorized part to be well above the rule’s $L = 14K$, so that rule is an empirical setting rather than a majorization. This is one reason we also report the parameter-free variant that computes $\lambda_{\max}$ exactly. Adding extrapolation to a cell with $\sigma > 0$ has no such guarantee and we treat it as a heuristic. Finally, the whole of (1) is non-convex, so all of this concerns the convex subproblem, not the alternation.

Limits of the evidence. The recovery checks are deliberately small (K up to 8, N up to 128^2 , M up to 32), because what they test is exactness against a reference, not performance; the colour runs are at the filter count, filter size, channel count and image size of the published colour experiment but with $K = 4$ training images rather than forty. All runs use one training-set seed and one initial-dictionary seed, so the comparisons between cells are paired but the variation across training sets is not measured. We report serial timings only: consensus and its block form are worth what they are worth because their replicas can run concurrently, and we do not measure that. The interior optimum of Section 6.1 is a measured property of these problem sizes, not a theorem, and the value of p that minimises time-to-target should be expected to move with $|X|$, M and N .

10 Related work

The dictionary updates we recover, their comparison, the observation that the “3D” method is consensus after a change of basis, the penalty-scaling analysis in K , and the statement that the channel count plays for sparse coding the role the signal count plays for the dictionary update, are all due to Garcia-Cardona and Wohlberg [2018a], building on Wohlberg [2016b] and Šorel and Šroubek [2016]. Our contribution is not those methods but the single derivation that generates them, what it says about which of them are the same, and what it contains that they do not. The multi-channel representations are also not ours: they are enumerated and compared by Wohlberg [2016a] and Garcia-Cardona and Wohlberg [2018b], the latter giving a taxonomy that includes a product dictionary we do not treat at all. Running the known updates on colour data is replication of that work, and we report it as the baseline it is; what is new here is that the two representations are one iteration with the channel index in different slots, the exact identity of Proposition 1, and the penalty rules that follow from it, which Garcia-Cardona and Wohlberg [2018b] leaves to a scaling analysis in C rather than to an identity in KC . Beyond those two representations the multi-channel literature adds structure we do not treat: rank-one multivariate atoms with a single shared activation [Dupré la Tour et al., 2018], fully three-dimensional spectral filters [Barajas-Solano et al., 2020], and cross-modality decompositions into common and unique features [Gao et al., 2022]. Each places the channel index in a way our row-index/right-hand-side dichotomy does not cover, and each is a candidate for the same treatment.

Two other framings deserve mention because they are the nearest things to this one. Bristow and Lucey [2014] put the convolutional sparse-coding literature into a common notation, but predate the consensus, iterated-Sherman–Morrison and proximal-gradient dictionary updates and do not generate them; and Rey-Otero et al. [2020] enumerate variations of the convolutional *model* — stride, sampling — where we enumerate variations of the solver for one subproblem of it. Two further axes are orthogonal to ours and could be composed with it: online and streaming CDL [Liu et al., 2018], which replaces the batch over signals by a surrogate and so changes what $\mathcal{I}$ contains; and penalty adaptation [Peng, 2019], which varies σ during the run rather than fixing it as we do.

Grouped consensus is standard in distributed optimisation [Pierra, 1984, Spingarn, 1985, Boyd et al., 2011] and has been used for convolutional sparse coding by Choudhury et al. [2017], who partition the stacked data into a free number of blocks with local dictionary copies coupled by averaging and projection — the same cell we report in Section 6, which is why we credit rather than claim it. Skau and Wohlberg [2018] group along the *filter* index for sparse coding, a different axis of the same idea. Moreau and Gramfort [2022] partition the spatial domain asynchronously and dispense with consensus averaging. On the majorization side, Chun and Fessler [2018] give a block-proximal-gradient framework with designed majorizers and momentum for CDL, with convergence guarantees; it is the closest existing framework, and it is complementary: its parameters are the block schedule and the majorizer, and it does not generate the equality-constrained, consensus, or frequency-domain updates. Silva and Rodríguez [2018] combine consensus with an accelerated proximal-gradient inner step, which in our terms is a partition value combined with a majorization value; Veshki and Vorobyov [2022] observe that existing methods differ mainly in how they treat one least-squares subproblem, which is the premise this paper formalises. Neither proposes a family in which the known updates are parameter values. Analyses of how the number of consensus blocks trades per-iteration cost against convergence rate exist for distributed quadratic problems [Shi et al., 2014, Ghadimi et al., 2015]; Choudhury et al. [2017] note the trade-off empirically for CSC. We are not aware of an analogue for the majorization interior, which is the cell we contribute.

11 Conclusion

The convolutional dictionary update has been a list of methods. It is better understood as one iteration with five parameters, three of which have interiors. Writing it that way costs nothing — the same code, instantiated by parameter values, reproduces every method on the list to machine precision — and it pays in three ways. It makes equivalences provable rather than observed: the multi-channel problem with a shared dictionary *is* the single-channel problem with KC signals, and a change of basis on the signal index set is a gauge exactly when it respects the partition. It makes the multi-channel design choice legible: the channel count inflates the rank of exactly one of the two subproblems, and the representation decides which, with the penalty rules following the rank. And it produces methods: keeping part of the rank exact and majorizing the rest is not on the list, collapses onto two different published methods at its two ends, and is faster than either in between.

The obvious next step is the subproblem we held fixed. Sparse coding has the same structure with the channel count in the role of the signal count; the same replicate-and-majorize construction should apply, and the partial-rank cell should be useful there exactly when the channel count is large — hyperspectral data, where the rank is in the hundreds. Whether the two stages should use matching parameter values, or whether the alternation is better served by mismatched ones, is a question this framework can ask and the separate derivations could not.

Reproducibility

All code, logs and figures are in the accompanying directory: the template and its driver in `code/template.py` and `code/driver.py`, one run script per experiment, and one JSON log per run in `logs/`; Appendix A records what was reused from the reference implementation and what was not. Every number in this paper is generated from those logs into `logs/numbers.tex`; the paper contains no transcribed values. The parameterisation, the cells we set out to recover, and the empty cell we set out to occupy were written down in `code/framework_plan.md` before the runs, with one amendment, dated and explained there and in `attempts_log.md`, when a literature check showed that the cell originally chosen was already taken.

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A Implementation and what was reused

The array layout, the constraint prox, the diagonal-plus-low-rank solvers (`solvemdbi_ism`, `solvemdbi_cg`, `solvedbi_sm`), the FFT wrappers and the sparse-coding solver are SPORCO’s [Wohlberg, 2017] and are used unmodified; `codebase_notes.md` records what was inspected and reused. The iteration (5)–(7), its parameterisation, the partition and basis machinery, and the per-block majorization are this work’s. The dictionary-update solver is a single class of about two hundred lines with one step method; there is no dispatch on method name anywhere in it.

Two implementation details affect exact reproduction of the reference runs. The reference implementation initialises the scaled dual variable of the equality-constrained solvers to the initial dictionary and that of the consensus solver to the initial dictionary divided by the penalty; we expose the initialisation as a scalar multiple of the initial dictionary and use each family’s own value when reproducing it. And the constraint prox for a multi-channel dictionary normalises each filter over its channels jointly, whereas the multi-channel functional as usually written asks for unit norm per channel; we follow the reference implementation, since the recovery targets are defined by it, and note that the two differ.

B Additional recovery results

Table 9 repeats the recovery at the larger setting.

C The interior cells in the multi-channel representation

Figure 9 is the partial-rank sweep of Section 6.1 in the single-channel-dictionary colour representation, and Figure 10 compares the two step policies on its extrapolated edge. Both are referred to in Section 6.1.

Table 9: Recovery at $K = 8$, $N = 128^2$, $M = 32$, 100 iterations, seed 777.

cell	functional deviation	dictionary deviation	converged functional
equality constraint, ISM	4.4×10^{-16}	4.4×10^{-15}	126.4824
equality constraint, CG	2.2×10^{-9}	6.3×10^{-8}	126.4992
consensus	4.4×10^{-16}	3.5×10^{-15}	127.2470
proximal gradient	4.6×10^{-16}	1.2×10^{-14}	126.4292

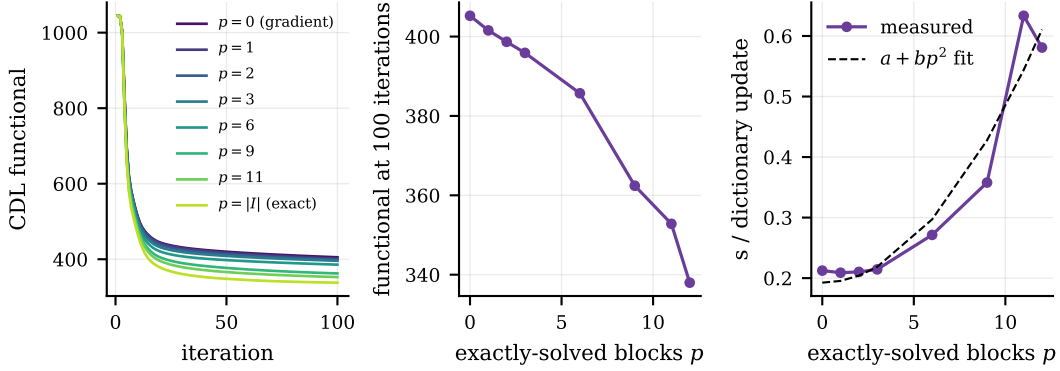


Figure 9: Partial-rank majorization in the single-channel-dictionary colour representation, where the signal index set is the $K \times C$ image-channel pairs ($|I| = 12$). Left: functional against iteration as p grows. Centre: functional after 100 iterations against p . Right: measured cost of one dictionary update.

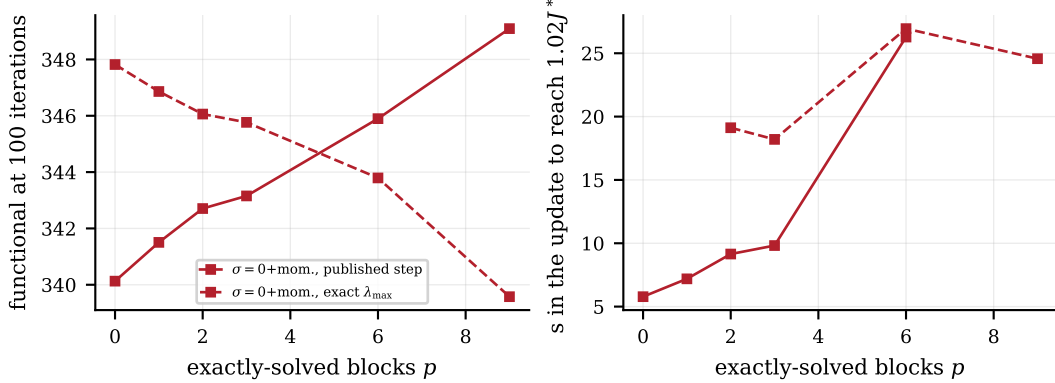


Figure 10: The majorization interior in the same representation, on the extrapolated edge under both step policies. With the parameter-free step the interior reaches a lower functional than the proximal-gradient corner (Section 6.1).

D Learned dictionaries

Figure 11 shows the filters the two multi-channel representations learn from the same colour images: a multi-channel dictionary whose filters carry colour, and a single-channel dictionary shared by the three channels.

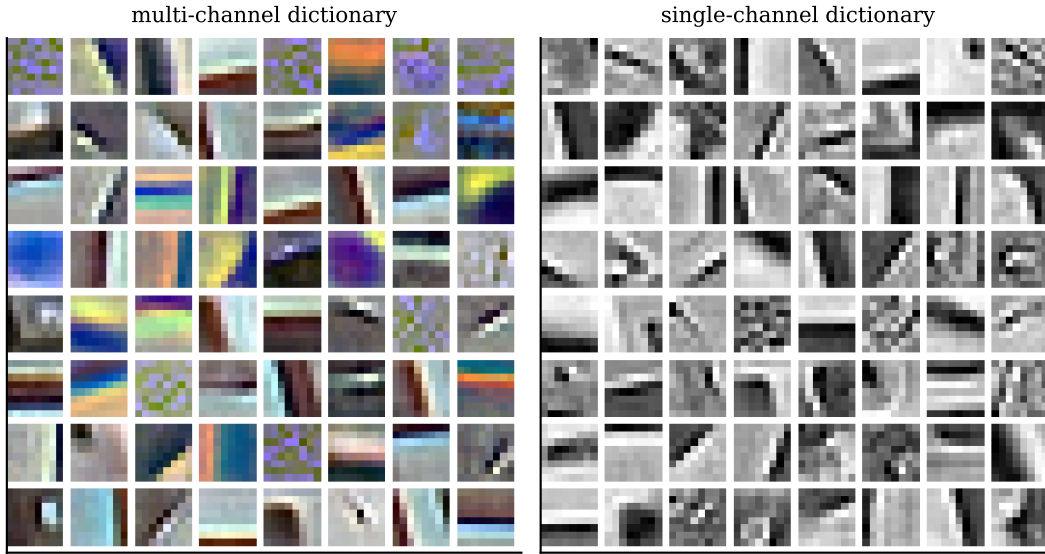


Figure 11: Dictionaries learned by the gradient instance in the two multi-channel representations, at the settings of Section 5.